Genetic drift, with and without mutation

Answer key — Journal club

N gene copies, mutation probability u each way per generation, then binomial resampling; x is the fraction of type A, t is in generations, $\theta = 2Nu$, and B is the variance per unit time. Equation numbers are Gardiner, 2nd edition.

Key results

forward equation:	$\partial_t p = -\partial_x\big[u(1-2x)\,p\big] + \tfrac{1}{2}\partial_x^2\Big[\dfrac{x(1-x)}{N}p\Big]$
stationary state:	$p_s = \dfrac{\Gamma(2\theta)}{\Gamma(\theta)^2}\big[x(1-x)\big]^{\theta-1}, \qquad H_s = \dfrac{\theta}{1+2\theta}$
the end $x=0$:	regular for $0<\theta<1$, entrance for $\theta \geq 1$, exit for $\theta = 0$; $d = 2\theta$
with mutation:	$\lambda_n = \dfrac{n(n-1+2\theta)}{2N}, \quad Q_1 = 1-2x, \quad Q_2 = x(1-x) - \dfrac{\theta}{2(2\theta+1)}; \quad P_n = p_s Q_n$
without mutation:	$\lambda_n = \dfrac{n(n-1)}{2N}; \quad Q_0 = 1,\ Q_1 = x; \quad P_n = \dfrac{Q_n}{x(1-x)}\ (n \geq 2)$
decaying modes:	$P_2 = 1,\ Q_2 = 6x(1-x); \qquad P_3 = 1-2x,\ Q_3 = 30x(1-x)(1-2x)$
QSD, survival:	uniform; $S \simeq 6x_0(1-x_0)\,e^{-t/N}; \quad \dfrac{p}{S} \simeq 1 + 5(1-2x_0)(1-2x)\,e^{-2t/N}$
absorption:	$\Pr(A \text{ takes over}) = x_0, \qquad \langle T\rangle_{\text{QSD}} = N = 1/\lambda_2$
mean exit time:	$T(x_0) = -2N\big[x_0 \ln x_0 + (1-x_0)\ln(1-x_0)\big], \quad T(\tfrac{1}{2}) = 2N\ln 2$
rare mutation:	$\dfrac{p_s Q_2}{\theta/2} \to 1 - \tfrac{1}{2}\delta(x) - \tfrac{1}{2}\delta(x-1), \qquad \lambda_1 = 2u = \text{twice the substitution rate}$

(a) Build the forward equation

Mutation takes x to $x' = x + u(1-2x)$. Resampling gives a new fraction with mean x' and variance $x'(1-x')/N$. So, exactly,

$$\langle \Delta x\rangle = u(1-2x), \qquad \langle (\Delta x)^2\rangle = \frac{x'(1-x')}{N} + u^2(1-2x)^2 = \frac{x(1-x)}{N} + O\Big(\frac{u}{N}, u^2\Big).$$

With $u = O(1/N)$ the corrections are $O(1/N^2)$, and so are the higher moments. Per generation, $A = u(1-2x)$ and $B = x(1-x)/N$, so

$$\partial_t p = -\partial_x\big[u(1-2x)\,p\big] + \tfrac{1}{2}\partial_x^2\Big[\frac{x(1-x)}{N}\,p\Big], \qquad \mathcal{L}^\dagger f = u(1-2x)\,f' + \frac{x(1-x)}{2N}\,f''.$$

- Mutation gives the drift, $A = -2u\,(x - \tfrac{1}{2})$, which pulls x toward $\tfrac{1}{2}$ at rate $2u$. Resampling is unbiased, $\langle x_{\text{new}}\rangle = x'$, so it contributes only the noise B.
- A population of one type resamples to that type: there is nothing to sample from, so $B(0) = B(1) = 0$. Mutation creates the missing type, so $A(0) = u > 0$ and $A(1) = -u < 0$ point inward.
- Both A and B are of order $1/N$, so x changes by order one in order N generations. The natural time unit is N generations, and on it only θ matters.

(b) Find the stationary state

1. $2A/B = 2Nu(1-2x)/[x(1-x)] = \theta[1/x - 1/(1-x)]$, which integrates to $\theta \ln[x(1-x)]$. Then (5.2.39) gives $p_s \propto B^{-1}[x(1-x)]^\theta \propto [x(1-x)]^{\theta-1}$, and the Beta integral normalises it:

$$\boxed{p_s(x) = \frac{\Gamma(2\theta)}{\Gamma(\theta)^2}\,[x(1-x)]^{\theta-1}}$$

Mutation pushes x toward $\frac{1}{2}$; resampling noise pushes it toward the ends, where it then lingers, because B is small there. For $\theta > 1$ mutation wins and p_s has a hump at $\frac{1}{2}$. At $\theta = 1$ it is flat. For $\theta < 1$ it is U-shaped but integrable: most of the time one type is nearly fixed, until a mutation that spreads by chance swaps them (figure, panel a).

2. With $x_0 = \frac{1}{2}$ in (5.2.25)–(5.2.28), $f = [4x(1-x)]^{-\theta}$ and $g = 2/(Bf) = 2N[x(1-x)]^{-1}[4x(1-x)]^{\theta}$. Near 0, $f \sim x^{-\theta}$ and $g \sim x^{\theta-1}$.

- $0 < \theta < 1$: f and g are both integrable, so $x = 0$ is **regular**.
- $\theta \geq 1$: f is not integrable, g is. Then $\int_x^{1/2} f \sim x^{1-\theta}$ (or $\ln(1/x)$ at $\theta = 1$), so $h_2 \sim$ const (or $\ln(1/x)$), which is integrable: **entrance**.
- For later, $\theta = 0$: $g \sim 1/x$ is not integrable, while $h_1 = f \int g \sim \ln(1/x)$ is: **exit**.

3. In the discrete model nothing passes beyond $x = 0$, and nothing stays there: a population of type a gains about $Nu = \theta/2$ new A copies per generation, so it leaves $x = 0$ within about $2/\theta$ generations, which is instantaneous on the time scale N. So $J(0) = 0$, the reflecting condition (5.2.13). A stationary current is the same everywhere, so $J = 0$ throughout, and the zero-current potential solution is the answer. A regular boundary would also allow stationary states with $J \neq 0$, for instance if populations at $x = 0$ were removed and restarted elsewhere. The model rules that out. For $\theta \geq 1$ there is no choice: an entrance boundary is never reached.

4. The quick rule sees $B(0) = 0$ and $A(0) = u > 0$ and says *entrance* for every $\theta > 0$. That is wrong for $\theta < 1$. The rule needs $\sqrt{B}$ to be Lipschitz, and $\sqrt{B} \simeq \sqrt{x/N}$ is not. Near 0, $A \simeq u$ and $B \simeq x/N$. Rescaling $y = 4DNx$ gives $\mathrm{d}y = 4DNu\,\mathrm{d}t + \sqrt{4Dy}\,\mathrm{d}W$: the squared radius without the spring, with $dD = 4DNu$, so

$$d = 4Nu = 2\theta.$$

The talk's thresholds $d = 2$ and $d = 0$ are $\theta = 1$ and $\theta = 0$. Near loss, the frequency of A behaves like the squared distance from the origin of a Brownian particle in 2θ dimensions, which returns to the origin only in fewer than two dimensions.

(c) Find the eigenfunctions

1. $\mathcal{L}^\dagger x^n = [nu + \frac{n(n-1)}{2N}]x^{n-1} - [2nu + \frac{n(n-1)}{2N}]x^n$. So $\mathcal{L}^\dagger$ is triangular on $1, x, \ldots, x^n$. Its eigenvalues are the diagonal entries, and the eigenvector for each is a polynomial of degree n (they are Jacobi polynomials in $1-2x$):

$$\boxed{\lambda_n = 2nu + \frac{n(n-1)}{2N} = \frac{n(n-1+2\theta)}{2N}}$$

2. $\mathcal{L}^\dagger(1-2x) = -2u(1-2x)$, so $Q_1 = 1-2x$ with $\lambda_1 = 2u$. Next, $\mathcal{L}^\dagger[x(1-x)] = u(1-2x)^2 - x(1-x)/N = u - \lambda_2\, x(1-x)$ with $\lambda_2 = 4u + 1/N$. So $\mathcal{L}^\dagger[x(1-x)+c] = -\lambda_2[x(1-x)+c]$ requires $c = -u/\lambda_2$:

$$Q_1 = 1-2x, \qquad Q_2 = x(1-x) - \frac{\theta}{2(2\theta+1)}.$$

3. $\langle Q_1(x(t))\rangle = Q_1(x_0)\,e^{-\lambda_1 t}$ exactly, which is $\langle x\rangle - \frac{1}{2} = (x_0 - \frac{1}{2})\,e^{-2ut}$. Resampling is unbiased, so it scatters x without moving its mean, and only mutation moves the mean. Since $\lambda_1 = 2u < \lambda_2$, the gap is the mutation rate itself, however large N is. The discrete chain gives $(1-2u)^t$ for the same quantity.

4. The stationary current is zero, so (5.2.71)–(5.2.76) apply; the boundary terms vanish because $Bp_s \sim x^\theta \to 0$. With $Q_0 = 1$, $\int P_2\,\mathrm{d}x = \int p_s Q_2\,\mathrm{d}x = \langle Q_2\rangle_s = 0$, so

$$\langle x(1-x)\rangle_s = \frac{\theta}{2(2\theta+1)}, \qquad H_s = \frac{\theta}{1+2\theta}.$$

This agrees with the Beta moment $\theta^2/[2\theta(2\theta+1)]$. Normalised, $P_1 = (2\theta+1)\,p_s\,(1-2x)$, since $\langle Q_1^2\rangle_s = 1/(2\theta+1)$. Because $x = \frac{1}{2}(Q_0 - Q_1)$, its stationary autocorrelation (5.2.90) is a single exponential: $\langle x(t)x(0)\rangle_s - \frac{1}{4} = e^{-2ut}/[4(2\theta+1)]$.

5. $\langle x(1-x)\rangle$ obeys the closed equation $\frac{\mathrm{d}}{\mathrm{d}t}\langle x(1-x)\rangle = u - \lambda_2\langle x(1-x)\rangle$. Resampling destroys diversity at rate $1/N$: two copies share a parent with probability $1/N$ per generation. Mutation alone would restore it at rate $4u$: $x(1-x) = \frac{1}{4}(1-Q_1^2)$, and Q_1^2 relaxes at $2\lambda_1$. Then $H_s = 2u/\lambda_2$ is the balance of the two, which is the result in 4 again.

(d) Switch off mutation

1. With $A = 0$ the potential solution is $\propto 1/B = N/[x(1-x)]$, which is not integrable at either end. So there is no stationary density. By (b) with $\theta = 0$, both ends are **exit** boundaries: they are reached in finite time and never left. One type is lost and the other fixed for good. The long-time state is $(1-\pi)\,\delta(x) + \pi\,\delta(x-1)$, which is not a density on $(0,1)$.

2. $\mathcal{L}^\dagger x = 0$, so $\langle x(t)\rangle = x_0$ for all t: the frequency is a martingale. As $t \to \infty$, $x \to 0$ or 1, so $x_0 = \langle x(\infty)\rangle = \Pr(A \text{ takes over})$. From (5.2.186)–(5.2.187), $\frac{1}{2}B\pi'' = 0$ with $\pi(0) = 0$, $\pi(1) = 1$, so $\pi = x_0$. Equivalently, $\psi = 1$ in the corrected (5.2.189)–(5.2.190). A single new mutant, $x_0 = 1/N$, takes over with probability $1/N$.

3. $\mathcal{L}^\dagger[x(1-x)] = -x(1-x)/N$, so $\langle x(1-x)\rangle = x_0(1-x_0)\,e^{-t/N}$: diversity is lost at $1/N$ per generation. If $\mathcal{L}^\dagger Q = \frac{x(1-x)}{2N}Q'' = -\lambda Q$ and $P = Q/[x(1-x)]$, then $BP = Q/N$, and

$$\tfrac{1}{2}\partial_x^2(BP) = \frac{Q''}{2N} = -\frac{\lambda Q}{x(1-x)} = -\lambda P.$$

The role of p_s is played by $1/[x(1-x)]$, the potential solution of 1, although it cannot be normalised. As in (5.2.91), which uses the reflecting p_s for absorbing walls, the relation $P = p_s Q$ outlives the stationary state. So $P_2 \propto 1$, and

the QSD is uniform on $(0,1)$, $\qquad P_2 = 1, \quad Q_2 = 6x(1-x), \quad S(t) \simeq 6x_0(1-x_0)\,e^{-t/N}$

from $\int_0^1 x(1-x)\,\mathrm{d}x = \frac{1}{6}$. Among populations that still carry both types, every frequency is equally likely, whatever x_0. The odd modes carry no probability, by symmetry, so the next correction to S decays at $\lambda_4 = 6/N$.

The atoms. $P_2 = 1$ on $(0,1)$ has total mass 1 and mean $\frac{1}{2}$. Biorthogonality with the zero modes needs $\int P_2 \cdot 1 = 0$ and $\int P_2\,x = 0$. The full forward mode therefore also has point masses at the ends, $P_2 = 1 - \frac{1}{2}\delta(x) - \frac{1}{2}\delta(x-1)$: what the interior loses, the ends gain, half each. Part (e) meets this object again.

4. For the QSD $p = 1$, $J = -\frac{1}{2}\partial_x(Bp) = -(1-2x)/2N$. So $J(0) = -1/2N$ flows out through 0 and $J(1) = +1/2N$ out through 1. The total is $1/N = \lambda_2$: a QSD loses probability at its own rate. Write $J = -\frac{1}{2}B'p - \frac{1}{2}Bp'$. Where $B > 0$, an absorbing wall needs $p = 0$ (5.2.14), and the loss is a diffusive flux $-\frac{1}{2}Bp'$. Here $B(0) = 0$ and $B'(0) = 1/N$, so $J(0) = -p(0)/2N$. The slope of the noise strength acts as a velocity $-\frac{1}{2}B'$ that carries probability into the end, and the loss is set by the density there, not its slope. An exit boundary imposes no condition, so nothing forces $p = 0$; the forward equation only requires p to be integrable.

5. $x(1-x)(1-2x) = x - 3x^2 + 2x^3$ has second derivative $-6(1-2x)$, so $\mathcal{L}^\dagger Q_3 = -(3/N)Q_3$ and $\lambda_3 = 3/N$. The partner is $P_3 = 1 - 2x$, and normalising gives $Q_3 = 30x(1-x)(1-2x)$. Since P_3 carries no probability,

$$\frac{p(x,t)}{S(t)} \simeq 1 + \frac{Q_3(x_0)}{Q_2(x_0)}\,P_3(x)\,e^{-(\lambda_3-\lambda_2)t} = 1 + 5(1-2x_0)(1-2x)\,e^{-2t/N}.$$

The approach rate is $\lambda_3 - \lambda_2 = 2/N$. From $x_0 = \frac{1}{2}$, $Q_3(x_0) = 0$ and all odd modes are absent, so the approach is at $\lambda_4 - \lambda_2 = 5/N$. Here $\lambda_3 - \lambda_2 = 2\lambda_2$: the conditioned density settles only a little faster than populations are absorbed, so there is no metastability. Panel (b) shows the formula at work: at $t = N$ from $x_0 = 0.1$, $p/S \simeq 1 + 0.54(1-2x)$.

6. With $A = 0$, (5.2.154) is $\frac{x(1-x)}{2N}T'' = -1$, so $T'' = -2N[1/x + 1/(1-x)]$ with $T(0) = T(1) = 0$:

$$\boxed{T(x_0) = -2N\big[x_0 \ln x_0 + (1-x_0)\ln(1-x_0)\big], \qquad T(\tfrac{1}{2}) = 2N\ln 2 \approx 1.39N}$$

A single mutant is usually lost fast: $T(1/N) \simeq 2(1 + \ln N)$ generations. Started from the QSD, the interior density stays uniform and shrinks as $e^{-\lambda_2 t}$ exactly. So the absorption time is exponential with mean $1/\lambda_2 = N$, and $\int_0^1 T\,\mathrm{d}x_0 = N$. Check: $\int_0^1 T\,\mathrm{d}x = -4N\int_0^1 x\ln x\,\mathrm{d}x = N$.

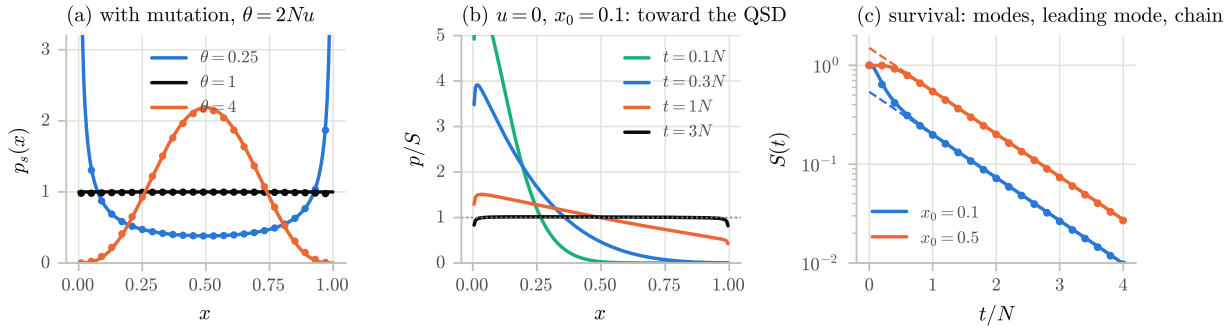


Figure 1: (a) The stationary density for $\theta = \frac{1}{4}$, 1, 4, with the exact stationary distribution of the discrete chain (dots, $N = 100$). (b) No mutation, $N = 200$, $x_0 = 0.1$: the chain's distribution, conditioned on both types surviving, flattens onto the uniform QSD. The dips at the edges are lattice effects, within a few copies of fixation. (c) Survival from $x_0 = 0.1$ and $\frac{1}{2}$: mode sum (solid), leading mode $6x_0(1-x_0)e^{-t/N}$ (dashed), and the chain (dots), $N = 200$.

(e) Optional: from one to the other

1. $\Gamma(2\theta)/\Gamma(\theta)^2 \simeq (1/2\theta)/(1/\theta)^2 = \theta/2$. The probability within ε of 0 is $\simeq \frac{\theta}{2}\cdot\varepsilon^\theta/\theta \to \frac{1}{2}$, while the density in the interior is $O(\theta)$. So $p_s \to \frac{1}{2}\delta(x) + \frac{1}{2}\delta(x-1)$: almost every population is fixed for one type, each half the time. And $\lambda_1 = 2u = \theta/N \to 0$, so $Q_1 = Q_0 - 2x$ becomes a combination of the

two zero modes of (d), 1 and the fixation probability x. Without mutation, every mixture of the two absorbed states is stationary; rare mutation picks the even one.

2. $p_s Q_2 = \frac{\Gamma(2\theta)}{\Gamma(\theta)^2}[x(1-x)]^\theta - \frac{\theta}{2(2\theta+1)}\,p_s$. The first term tends to $\frac{\theta}{2}$ on all of $(0,1)$. The second tends to $-\frac{\theta}{2}\big[\frac{1}{2}\delta(x) + \frac{1}{2}\delta(x-1)\big]$. Dividing by $\theta/2$ gives $1 - \frac{1}{2}\delta(x) - \frac{1}{2}\delta(x-1)$, which is the full P_2 of (d): the uniform QSD inside, and the probability it loses, deposited half at each end. It is orthogonal to $Q_0 = 1$ (mass $1 - \frac{1}{2} - \frac{1}{2} = 0$) and to $Q_1 = x$ ($\frac{1}{2} - \frac{1}{2} = 0$). Its rate also matches: $\lambda_2 = 4u + 1/N \to 1/N$.

3. From a population of type a, new A mutants arise at rate Nu per generation. Each starts at $x_0 = 1/N$ and takes over with probability $1/N$. So the population switches $a \to A$ at rate $Nu \cdot \frac{1}{N} = u$, and $A \to a$ at the same rate. A two-state process with rates u each way relaxes at $u + u = 2u = \lambda_1$, which is the exact gap from (c). The substitution rate equals the mutation rate, independent of N: Kimura's neutral rate. It is the double well of the talk, with the two fixed states as the wells and the gap $\lambda_1 \simeq 2r$. The picture needs the time between substitutions, $1/u$, to be much longer than a takeover, which lasts of order N generations, so $\theta \ll 1$. Equivalently $\lambda_1 = 2u \ll \lambda_2 \simeq 1/N$. For $\theta \gtrsim 1$ both types are usually present (panel a, $\theta = 1$ and 4), and the picture fails, though $\lambda_1 = 2u$ remains exact.

For the discussion

- **Two ways to lose a stationary state.** In the talk, the stationary state disappeared when the walls were made absorbing. Here no wall changes: switching off mutation turns both ends from regular or entrance into exit boundaries. The equation itself decides.
- **Survivors never fix.** Weighting paths by their chance of surviving forever (Doob's h-transform, with $h = Q_2$) adds the drift $B\,\partial_x \ln Q_2 = (1-2x)/N$. That is Wright–Fisher with mutation at $\theta = 2$. Its stationary density, $\propto x(1-x) = P_2 Q_2$, is Beta$(2,2)$, and its ends are entrance boundaries ($\theta \geq 1$), as they must be.
- **Kimura's solution.** For $u = 0$, the interior modes are $P_n = C_{n-2}^{(3/2)}(1-2x)$, Gegenbauer polynomials, and $Q_n \propto x(1-x)P_n$. Kimura (1955) found the full transition density this way.
- **Checks.** `scripts/wright_fisher.py` builds the transition matrix of the discrete chain. Its eigenvalues are exactly $(1-2u)^k \prod_{i<k}(1 - i/N)$, whose rates tend to λ_k. The script checks p_s, H_s and Feller's test, the residuals of every Q_n and P_n above, and biorthogonality. It compares survival from the mode sum with finite differences and with the chain, the fixation probability x_0, $T(\frac{1}{2})$, $\langle T \rangle_{\mathrm{QSD}} = N$, and the limits in (e).